

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number: Group 2

Start date: 29/12/2025

Names:

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Internship Duration: 8 Weeks

1. Project Title

URBAN PULSE - Smart City Mobility Intelligence Platform

2. Project Objective

The objective of the Infosys Urban Pulse project is to design and develop an interactive Power BI-based Smart City Mobility Intelligence Platform that brings together transportation, weather, and ride-hailing data on a single analytical platform.

The platform is intended to analyze urban mobility patterns, travel demand, revenue performance, operational efficiency, accessibility, seasonal trends, and mobility equality across different locations. By combining these data sources, the project aims to convert raw, disconnected urban data into meaningful, decision-ready insights that support smarter and more informed urban planning.

3. Project description in detail

This project focuses on building an end-to-end data analytics and business intelligence solution for smart-city mobility. It follows the complete analytics lifecycle — starting from data collection and preprocessing, moving through data integration and ETL (Extract, Transform, Load), and continuing into data modeling, DAX-based calculations, visualization, dashboard development, and final insight generation.

The platform analyzes transportation, weather, and ride-hailing information together to understand mobility demand, travel patterns, the influence of weather on travel behavior, operational performance, customer and vehicle behavior, revenue trends, seasonal variation, mode substitution, and mobility equality across different areas of the city.

The solution is implemented entirely in Microsoft Power BI and makes use of Power Query for data transformation, Star Schema and Snowflake Schema data modeling, DAX measures and calculated columns, KPI cards, and a range of visuals including maps, heat maps, Sankey diagrams, and other analytical charts.

4. Timeline Overview

Week	Activities Planned	Activities Completed
week 1	Data Visualization & Fundamentals	Learned visualization principles, data types, and data-to-decision flow.
week 2	Smart City Data & Integration	Explored datasets, data sources, integration challenges, and solutions.
week 3	ETL Process	Performed data extraction, transformation, cleaning, and loading.
week 4	Power BI Visualization	Created various charts, maps, heat maps, Sankey diagrams, and treemaps.
week 5	Data Types & Date/Time	Worked with data types, date/time fields, time slots, and peak-hour analysis.
week 6	Power Query & M Language	Performed preprocessing, grouping, and aggregation using Power Query and M..
week 7	Data Modeling	Built Star Schema, created fact/dimension tables, and established relationships.
week 8	DAX & Analytics	Created measures, calculated columns, KPIs, and analytical calculations.

5a. Key Milestones

Milestone	Description	Date Achieved
Milestone 1 – Initial Data Visualization	Visualization of transportation, weather, and ride-hailing datasets to understand their structure, patterns, and initial trends.	Week 1–2
Milestone 2 – Data Processing, Modeling & Dashboard Development	Data processing and cleaning, implementation of Star Schema and Snowflake Schema data models, and development of the initial interactive dashboards.	Week 3–5
Milestone 3 – Weather Analytics	Development of a dedicated Weather Analytics dashboard to analyze weather conditions and their impact on urban mobility.	Week 6
Milestone 4 – Advanced Mobility Analytics	Development of Ride-Hailing Mobility, Seasonal, Mode Substitution Analysis, and Mobility Equality dashboards.	Week 7–8

5b. Project execution Details

The project was executed as an end-to-end workflow, with each stage feeding into the next to progressively build the final mobility intelligence platform.

Data Collection → Data Understanding → Data Cleaning → Data Integration → ETL → Feature Engineering → Data Modeling → Star Schema → Snowflake Schema → DAX Measures → KPI Development → Data Visualization → Interactive Dashboard Development → Analytical Insights → Final Mobility Intelligence Platform

Data import from transportation, weather, and ride-hailing sources

Power Query preprocessing and data transformation

Missing-value handling and duplicate removal

Data-type conversion and date/time transformation

Feature engineering for analytical readiness

Defining relationships between fact and dimension tables

Building Star Schema and Snowflake Schema models

Creating DAX measures and calculated columns

Designing KPI cards, slicers, and filters

Building interactive charts, maps, and geographical visuals

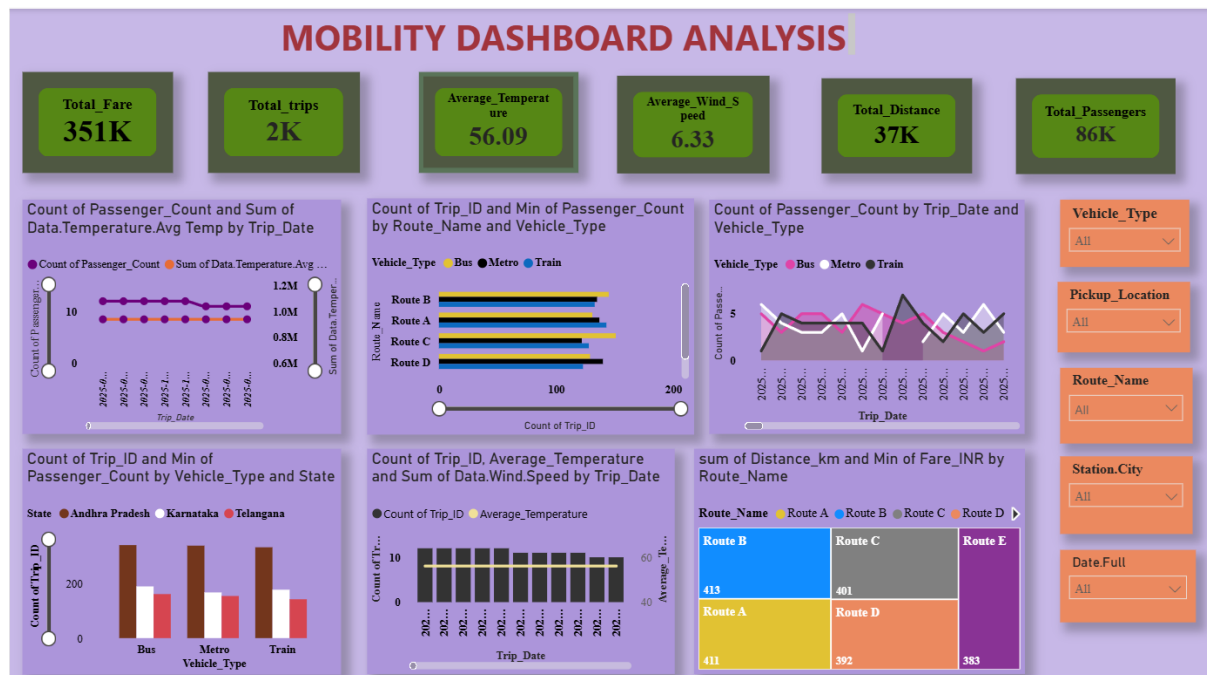
Generating dashboard-based analytical insights

6. Screenshots of Dashboards

Dashboard development formed the core deliverable of the project. Each dashboard was designed to answer specific analytical questions using a combination of KPI cards, charts, maps, and interactive filters, allowing users to explore mobility patterns from different perspectives.

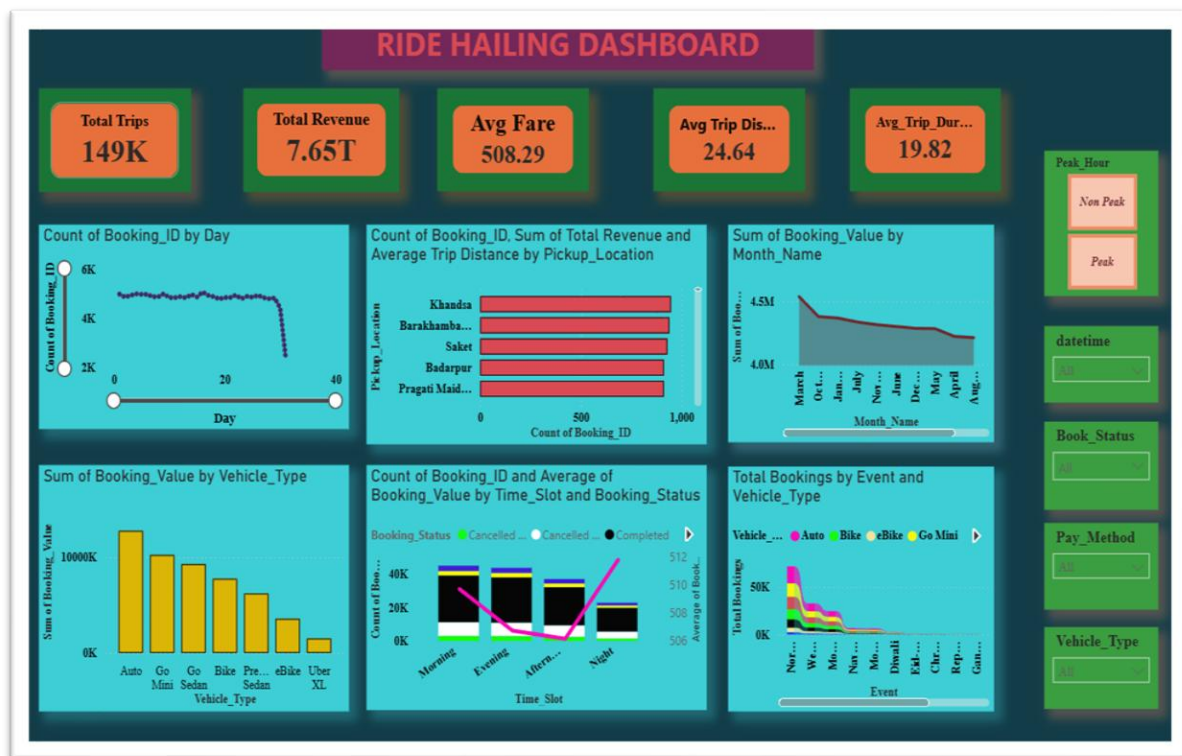
MOBILITY OVERVIEW DASHBOARD:

Provides a consolidated view of transportation and ride-hailing activity, including demand trends, peak time slots.

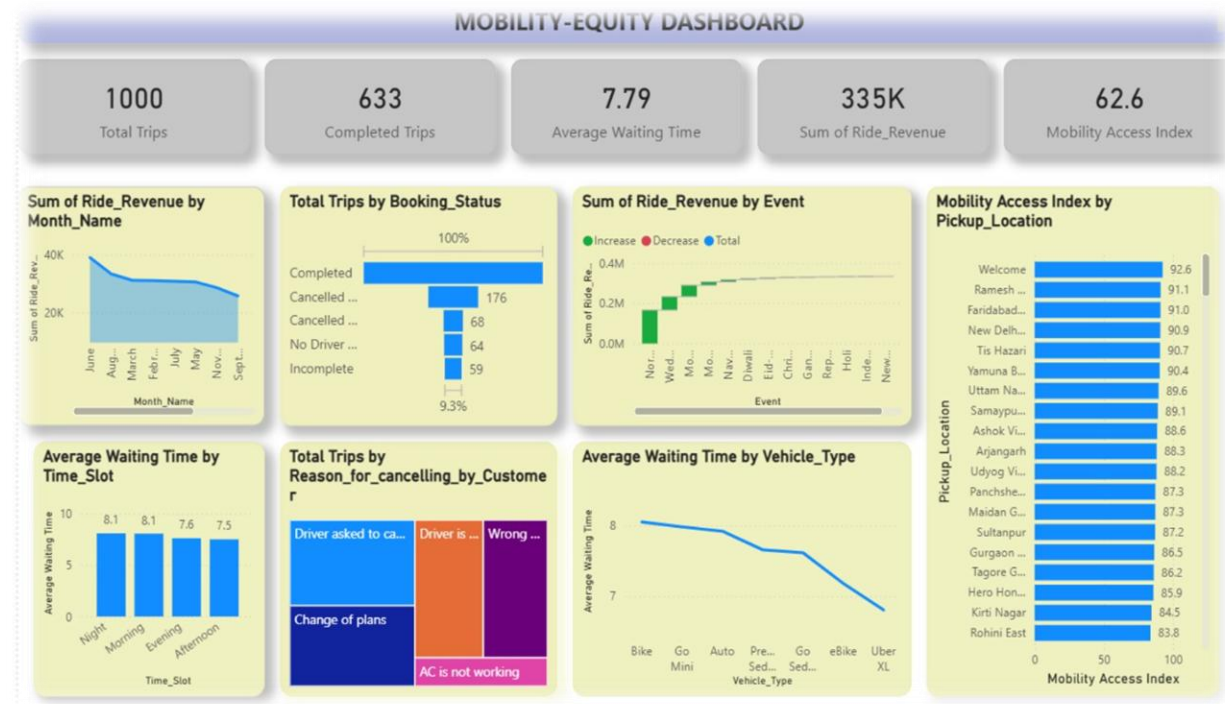


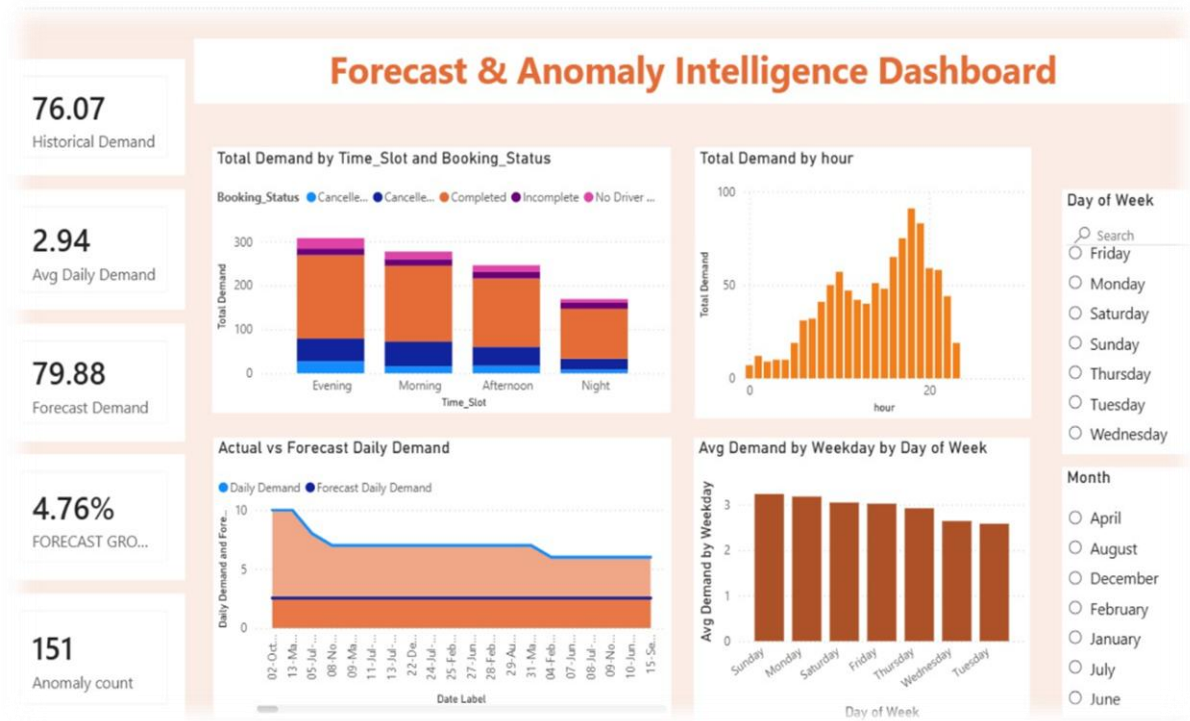
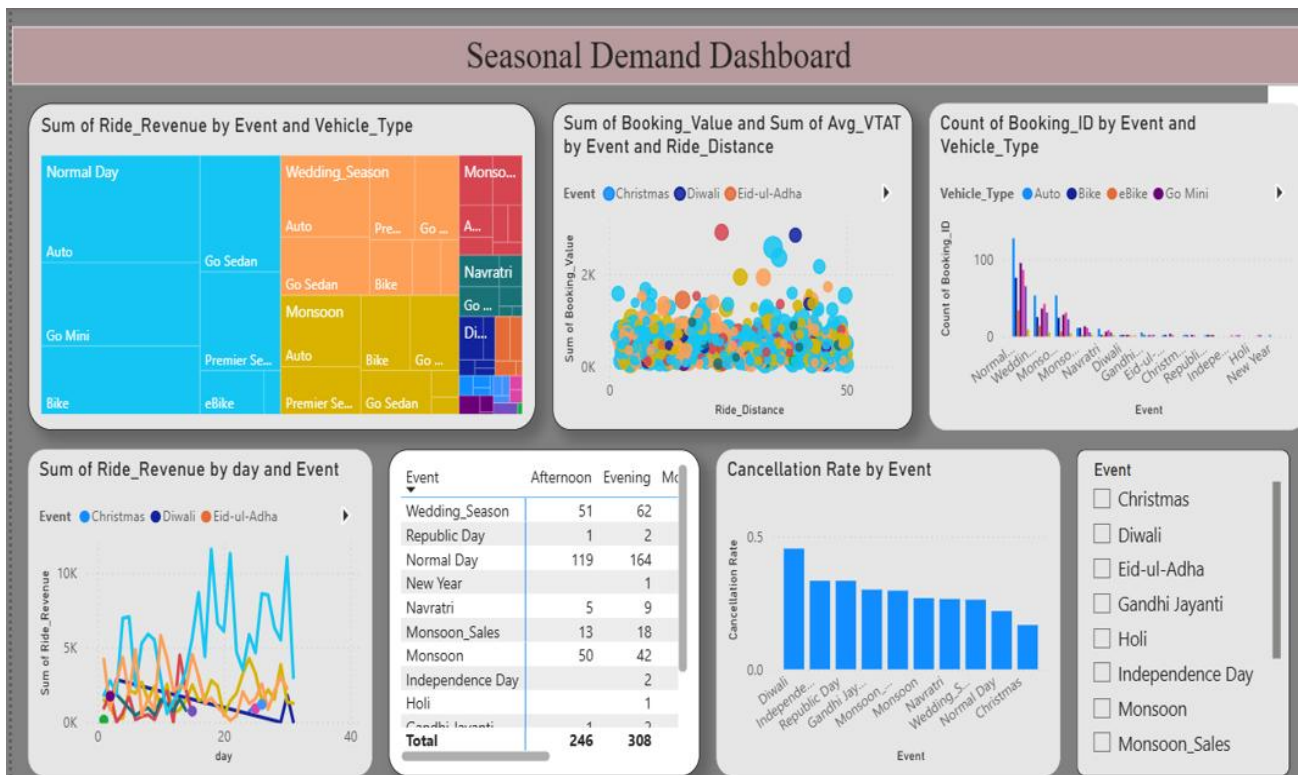
RIDE-HAILING MOBILITY DASHBOARD:

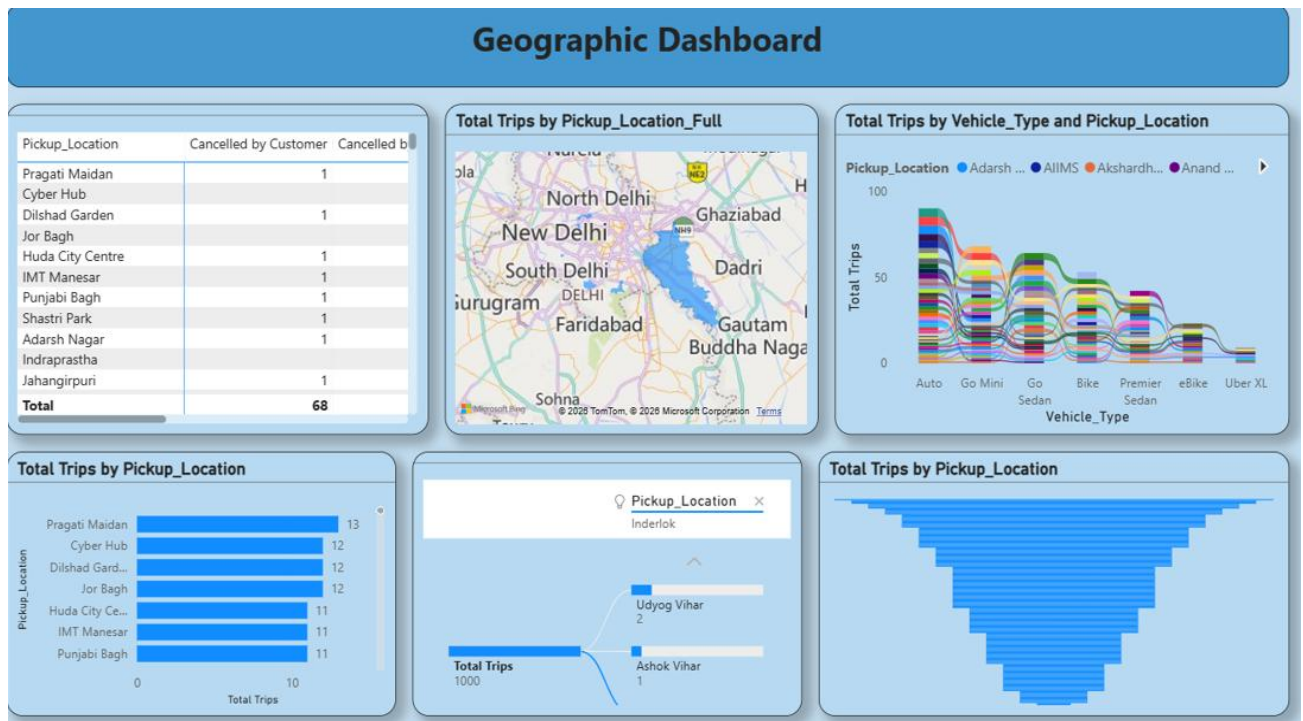
Focuses on ride demand, revenue, customer behaviour, and vehicle-level operational performance, using interactive charts and KPI indicators.

**MOBILITY EQUALITY DASHBOARD:**

Examines seasonal variation in mobility demand, mode substitution behavior, and mobility equality across different locations, using maps, heat maps, and Sankey diagrams



FORECAST AND ANAMOLY INTELEGENCE DASHBOARD:**Seasonal Demand Dashboard:**

Geographic Dashboard:**7. Challenges Faced**

Challenge	Approach Taken
Missing and inconsistent values	Identified and treated during the data cleaning stage using Power Query.
Duplicate records	Detected and removed to maintain data accuracy.
Different data formats	Standardized during preprocessing for consistency across datasets.
Data-type inconsistencies	Corrected through explicit data-type conversion in Power Query.
Integrating multiple datasets	Addressed through careful key mapping and relationship design.
Handling date and time fields	Managed using dedicated date tables and time-based transformations.
Designing appropriate relationships	Resolved through iterative testing of the data model.
Choosing Star vs. Snowflake Schema	Evaluated based on query performance and reporting needs.
Creating meaningful DAX measures	Refined through iterative testing against business questions.
Large numbers of categorical values	Managed using grouping and aggregation techniques.
Selecting appropriate visualizations	Guided by the nature of each analytical question.
Representing geographical mobility data	Addressed using Power BI maps and heat maps.
Unavailable custom visuals	Substituted with suitable native Power BI visuals.
Maintaining readable dashboards	Achieved through consistent layout and design standards.

8. Learnings & Skills Acquired

Working on this project provided the team with practical, hands-on exposure to the complete business intelligence lifecycle. Key learning outcomes include:

Data cleaning and preprocessing techniques

The end-to-end ETL workflow

Power Query and the M language

Integrating data from multiple sources

Power BI visualization techniques

Data modeling concepts, including Star and Snowflake schemas

Designing relationships and date tables

DAX, measures, and calculated columns

KPI creation and dashboard design

Geographical analysis using maps and heat maps

Generating business and analytical insights

Supporting data-driven decision making

Pros

End-to-end data analytics workflow, from raw data to final insights

Interactive and user-friendly Power BI dashboards

Multiple datasets integrated into a single analytical platform

Structured data modeling using Star and Snowflake schemas

Dynamic KPIs and DAX-based measures

Use of multiple visualization techniques for different insights

Strong mobility and geographical analysis capability

Supports data-driven decision making for smart-city planning

Cons

Dataset availability and quality limitations

Missing values present in some attributes

Limited real-time data; analysis is largely historical

Limited geographical coverage in the available datasets

Some advanced analysis is constrained by available dataset attributes

Overall results depend on the quality and completeness of source datasets

9. Testimonials from team

- Before this project, Power BI was just a name I had heard about. Working through the phases from Power Query to DAX helped me actually understand how a raw dataset turns into a dashboard someone can make decisions from.
- The most valuable part for me was data cleaning. Real datasets are messy, and figuring out how to handle missing values and inconsistent formats taught me more than any tutorial could.
- Working with three different datasets at once was challenging at first. Learning how to connect them through proper relationships and a Star Schema made everything click into place.
- I spent most of my time on the DAX measures. It took a few attempts to get the calculations right, but seeing the KPIs update correctly on the dashboard made it worth it.
- This project gave me a much better sense of how teams actually work together on a data project dividing tasks, reviewing each other's dashboards, and putting the final story together.

10. Conclusion

The Infosys Urban Pulse project successfully demonstrates an end-to-end Smart City Mobility Intelligence solution built using Microsoft Power BI. The project transformed raw transportation, weather, and ride-hailing datasets into structured, analysis-ready data, and converted them into interactive dashboards and analytical insights.

The final solution provides a consolidated understanding of the following aspects of urban mobility:

- Mobility demand
- Ride-hailing performance
- Revenue
- Customer behaviour
- Vehicle performance
- Weather patterns
- Seasonal trends
- Model substitution
- Mobility accessibility
- Mobility equality

Overall, the project demonstrates how data analytics and visualization can support a better understanding of urban mobility and assist data-driven smart-city decision making.

11. Acknowledgements

We would like to express our sincere gratitude to our Faculty/Mentor for their guidance, valuable feedback, and continuous support throughout the course of this project.

We are thankful to our Institution/Department for providing the academic platform, resources, and environment necessary to carry out this work.

We would also like to acknowledge our project team members for their collaboration, effort, and contribution at every stage of the project.

We extend our appreciation to the providers of the transportation, weather, and ride-hailing datasets and open-data sources used in this project.

Finally, we acknowledge Microsoft Power BI and the related tools that made the development of this analytics platform possible.